

Manual GSP Mining

*Sequential Patterns, Threshold Sensitivity
& a Comparison with Apriori*

⟨ A, B, D, E ⟩

⟨ B, C ⟩

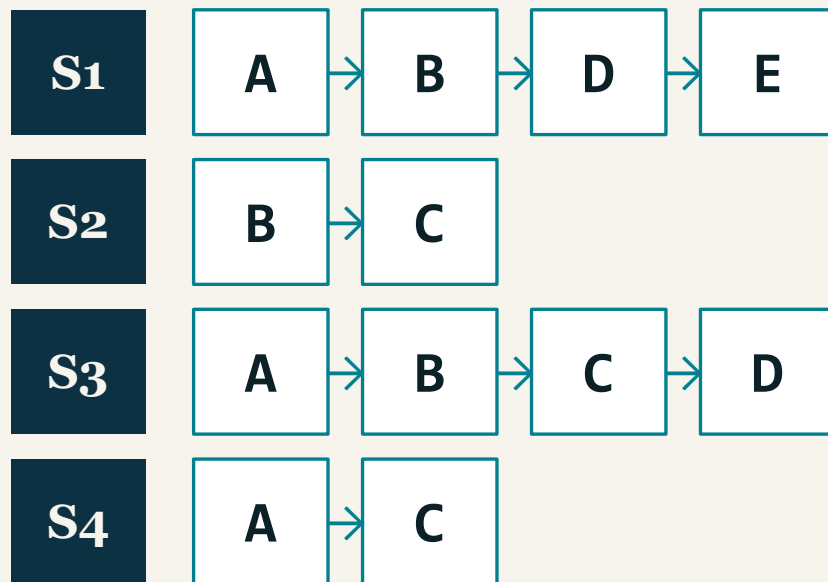
⟨ A, B, C, D ⟩

⟨ A, C ⟩

THE PROBLEM

Four customer journeys. Find every frequent ordered pattern.

DATASET



CONVENTION (TEXTBOOK GSP)

$T = \langle t_1, \dots, t_m \rangle$ is contained in $S = \langle s_1, \dots, s_n \rangle$
if there exist $i_1 < i_2 < \dots < i_m$ with $t_k = s_{i_k}$.

Order matters · items need not be adjacent.

Support of P = # input sequences containing P as a subsequence.



The GSP Loop

Four steps, repeated until no candidate survives.

01

Scan

Count support of every single item. Keep those $\geq \text{min_sup} \rightarrow L_1$.

02

Join

From L_k generate C_{k+1} via $\langle a,b \rangle \bowtie \langle b,c \rangle \rightarrow \langle a,b,c \rangle$.

03

Prune

Drop any candidate whose length- k sub-sequence is not in L_k .

04

Count

Scan the data once more. Keep candidates with support $\geq \text{min_sup}$.



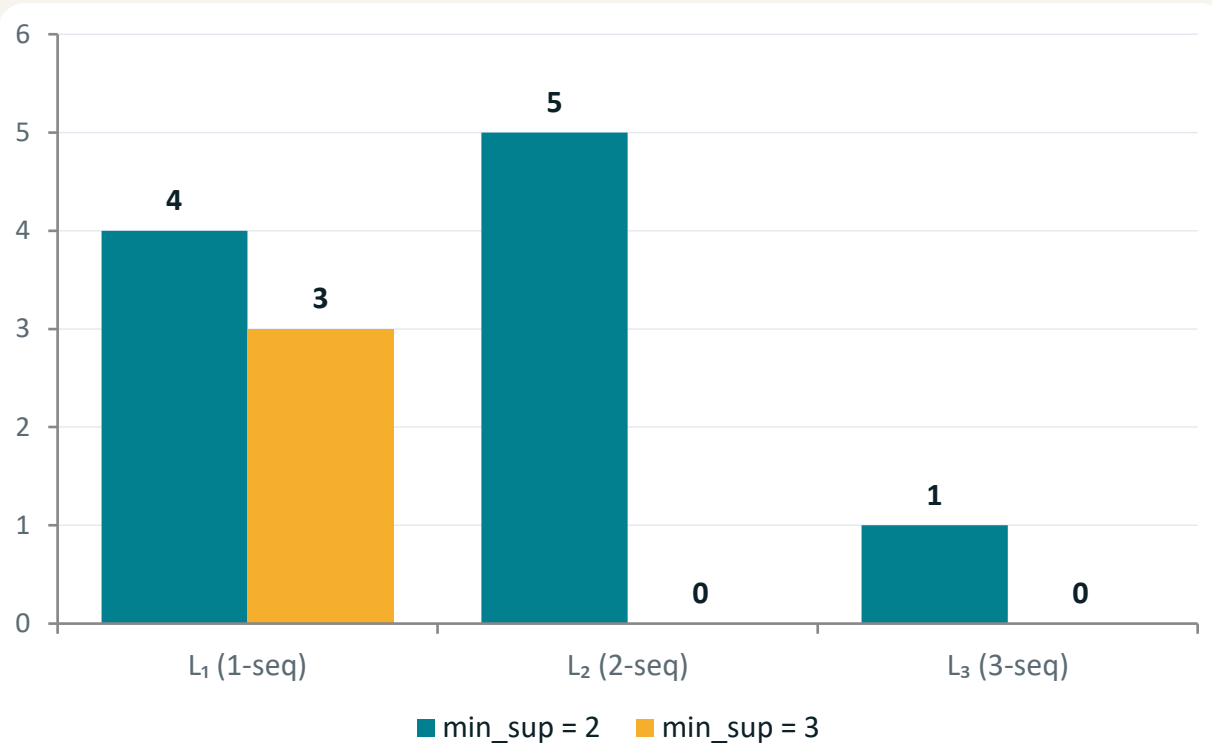
TASK 1 · min_support = 2

Ten frequent patterns across three levels.

L1	4 patterns	L2	5 patterns	L3	1 pattern
< A > < B > < C > < D >	sup 3 sup 3 sup 3 sup 2	< A, B > < A, C > < A, D > < B, C > < B, D >	sup 2 sup 2 sup 2 sup 2 sup 2	< A, B, D >	sup 2
< E > pruned (sup 1)		< C, D > pruned (sup 1) · all reverse pairs sup 0		< A, B, C > pruned (sup 1)	
WORTH NOTING Every frequent 2-pattern has support exactly 2 · all patterns sit right at the threshold.					

TASK 2 + TASK 3 · Raising the threshold

min_sup = 3 collapses the pattern set from 10 to 3.



Pattern count by level

DROPS OUT

⟨D⟩ from L₁, all five 2-seqs, and ⟨A,B,D⟩.

WHY

min_sup = 3 means a pattern must appear in 75 % of customers — strict at n = 4.

INSIGHT

A, B, C are universal touch-points. No multi-step journey is shared by ≥ 75 %.

-70 %

drop in pattern count
10 → 3 when raising
min_sup from 2 to 3

$\langle A, B, D \rangle$

Why this one — and not $\langle A, B, C \rangle$ or $\langle B, C, D \rangle$?

S1	A	B	D	E
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✓ $A \rightarrow B \rightarrow D$ found

S3	A	B	C	D
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✓ $A \rightarrow B \rightarrow D$ found (C skipped)

S2	B	C
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✗ no A

S4	A	C
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✗ missing B, D

Support = 2 · meets the threshold · no longer pattern can extend it (D never followed by another frequent item)

Same counts here — but only by coincidence.

GSP (sequences)

$L_1(4)$ $\langle A \rangle$ $\langle B \rangle$ $\langle C \rangle$ $\langle D \rangle$

$L_2(5)$ $\langle A, B \rangle$ $\langle A, C \rangle$ $\langle A, D \rangle$ $\langle B, C \rangle$ $\langle B, D \rangle$

$L_3(1)$ $\langle A, B, D \rangle$

Order preserved. $\langle X, Y \rangle \neq \langle Y, X \rangle$.

Apriori (itemsets)

$L_1(4)$ $\{A\}$ $\{B\}$ $\{C\}$ $\{D\}$

$L_2(5)$ $\{A, B\}$ $\{A, C\}$ $\{A, D\}$ $\{B, C\}$ $\{B, D\}$

$L_3(1)$ $\{A, B, D\}$

Order discarded. $\{X, Y\} = \{Y, X\}$.

WHY THEY MATCH HERE

Every co-occurring pair appears in only one order (alphabetical). No support is split between $\langle X, Y \rangle$ and $\langle Y, X \rangle$.

WHEN THEY DIVERGE

Add $S_5 = \langle D, B, A \rangle$: Apriori sees $\{A, B\}$ sup = 3 $\rightarrow$ frequent at min_sup=3; GSP splits into $\langle A, B \rangle=2$ and $\langle B, A \rangle=1 \rightarrow$ both fail.

Three lessons from four sequences.

01

Threshold sensitivity is non-linear

Raising min_sup from 2 $\rightarrow$ 3 cut the pattern count by 70%. The anti-monotone property turns a small threshold change into a cascading collapse through every level.

02

Order is information

GSP carries strictly more information than Apriori. On this dataset they agree; on any data with backward or repeated orderings they will not.

03

Pick the model that matches the data semantics

GSP for clickstreams, journeys, event logs. Apriori for market-basket co-occurrence.